$$\frac{1}{M}\sum_{i=1}^{m} \frac{-a^{(i)}}{\hat{a}^{(i)}} \frac{d\, a^{(i)}}{d w_j} + \frac{1-a^{(i)}}{1-\hat{a}^{(i)}} \frac{-d\, a^{i}}{dw}$$

$$\frac{d\,\hat{a}^{(i)}}{d w_j} = \frac{d}{dw_j} \frac{-1}{1+e^{-z^{(i)}}}$$

$$\frac{1}{M}\sum_{i=1}^{m}\left[-a^{(i)}(1-\hat{a}^{(i)}) + (1-a^{(i)})\,\hat{a}^{(i)}\right] x_j^{(i)}$$

$$\frac{1}{M}\sum_{i=1}^{m}(a^{(i)}-\hat{a}^{(i)})\, x_j^{(i)}$$

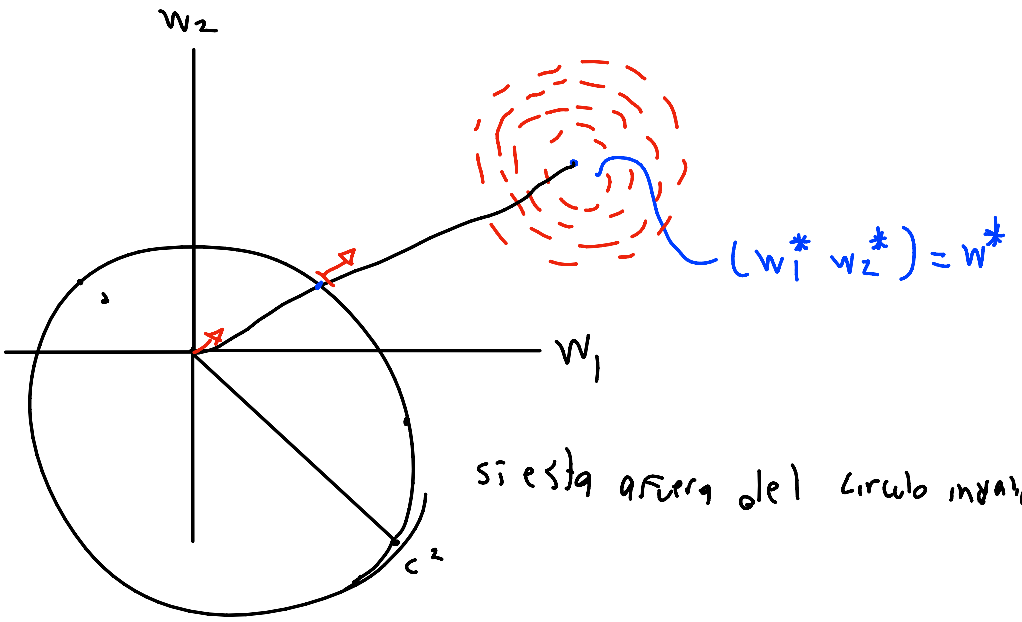
$$W \leftarrow W - \frac{dr}{M} X^T (A-\hat{A})$$

$$b \leftarrow b - \frac{dr}{M}\sum_{i=1}^{m}\left(a^{(i)} - \hat{a}^{(i)}\right)$$

$$W^*, b^* = \arg\min E_{in}(w,b)$$

bajo

$$\sum_{j=1}^{n} W_j^2 \leq C$$



si esta afuera del circulo invalido

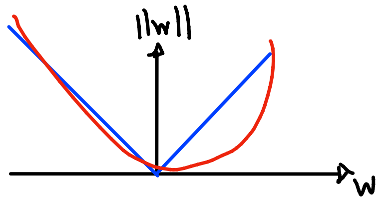
multiplicador de la granch

$$w^*_{reg}\, b^*_{reg} = \arg\min\left[E_{in}(w,b) + \frac{\lambda}{M}\sum_{j=1}^{n} w_j^2\right]$$

$$\frac{d}{dw_j}\left[E_{in}(w,b) + \frac{\lambda}{M}\sum_{j=1}^{n}\right] = -\frac{1}{M}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right) x_j^{(i)} - \frac{2\lambda}{M} w_j$$

$$w^*_{reg}\, b^*_{reg} = \arg\min\left[E_{in}(w,b) + \frac{\lambda}{M}\,\text{regu}(M)\right]$$

regu(w) =
- $\sum_{i=1}^{n} w_j^2 = ||w||^2$ — l_2
- $\sum_{i=1}^{n} |w_j|$ — l_1



reg lineal
- l_2 - Ridge
- l_1 - LASSO
- $l_1 + l_2$ — Elastic Net